

Two Random-Forcing Questions from arXiv:1909.09387 Answered by arXiv:2110.04568

Literature identification note

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Abstract

Two questions in Sobota and Zdomsky's *Convergence of measures in forcing extensions* are explicitly answered by their later paper *Convergence of measures after adding a real*. Random forcing destroys both the Nikodym and Grothendieck properties of every infinite ground-model Boolean algebra, and the random model satisfies $\mathfrak{d} < \mathfrak{ni}\mathfrak{k} = \mathfrak{gr}$. These results settle source Questions 6.13 and 6.4, respectively.

Identification status

- Run `fa_banach_001`, lane 13.
- Source: arXiv:1909.09387 [1].
- Answer: arXiv:2110.04568 [2].
- Classification: explicit later-literature answer; no new theorem claimed.

1 Random forcing and the Vitali–Hahn–Saks property

Source question (Question 6.13 of [1], PDF page 20). *Let $\mathbb{P}$ be random forcing and let $\mathcal{A}$ be a ground-model Boolean algebra which is σ -complete, or merely has property (I). Must $\mathcal{A}$ have the Vitali–Hahn–Saks property in the generic extension?*

Later theorem (Theorem 4.6 of [2], PDF page 10). *If a forcing adds a random real, then every infinite ground-model Boolean algebra has neither the Nikodym property nor the Grothendieck property in the generic extension.*

The answer to Question 6.13 is therefore *no*, in the stronger form that neither σ -completeness nor property (I) is relevant. Indeed, the Vitali–Hahn–Saks property implies both the Nikodym and Grothendieck properties, whereas the later theorem destroys both for every infinite ground-model algebra.

2 The cardinal invariants above $\mathfrak{d}$

Source question (Question 6.4 of [1], PDF page 18). *For $\mathfrak{x} \in \{\mathfrak{ni}\mathfrak{k}, \mathfrak{gr}, \mathfrak{vhs}\}$, is it consistent that $\mathfrak{x} > \mathfrak{d}$? In particular, is there an ω^ω -bounding forcing after which $\mathcal{P}(\omega)^V$ loses the Vitali–Hahn–Saks property?*

Corollary 5.3 of [2] (PDF page 20) states

$$\mathfrak{d} < \mathfrak{ni}\mathfrak{k} = \mathfrak{gr} \quad \text{in the random model.}$$

The source records $\mathfrak{ni}\mathfrak{k}, \mathfrak{gr} \leq \mathfrak{v}\mathfrak{h}\mathfrak{s}$, so this same model has $\mathfrak{d} < \mathfrak{v}\mathfrak{h}\mathfrak{s}$ as well. Thus Question 6.4 has a positive answer for every listed $\mathfrak{x}$. Moreover, measure-algebra random forcing is ω^ω -bounding, and Theorem 4.6 applied to the infinite algebra $\mathcal{P}(\omega)^V$ supplies exactly the forcing requested in the particular clause.

Scope

The later paper is by the same authors and explicitly describes the random case as addressing the preservation issue left open in the earlier work. This identification settles Questions 6.4 and 6.13 only; the other questions in Section 6 of arXiv:1909.09387 are not classified here.

References

- [1] D. Sobota and L. Zdomskyy, *Convergence of measures in forcing extensions*, arXiv:1909.09387 (2019).
- [2] D. Sobota and L. Zdomskyy, *Convergence of measures after adding a real*, arXiv:2110.04568 (2021; published 2024).